

KRISHNA Koushik Thokala

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Summary

Cloud Infrastructure Engineer building and validating AWS platforms end to end: multi-tier VPCs and managed EKS with Terraform, GitOps delivery through Argo CD and GitHub OIDC, and Prometheus burn-rate alerting against a 99.9% SLO, tested through autoscaling and failure-recovery drills. AWS Certified Solutions Architect – Associate; MS in Computer Science, Indiana University. Brings two years of production experience leading five-engineer shifts in a 24/7 carrier NOC.

Technical Skills

Cloud & Infrastructure as Code: AWS, Terraform, VPC, EC2, EKS, Lambda, RDS, DynamoDB, S3, API Gateway
Containers & Delivery: Kubernetes, Helm, Docker, ECR, Argo CD, GitOps, GitHub Actions, CI/CD, kubeconform
Security & Access: IAM, OIDC, IRSA, RBAC, KMS, Secrets Manager, Network Policies, Checkov
Observability & Reliability: Prometheus, Grafana, CloudWatch, k6, HPA, SLIs/SLOs, error budgets, burn-rate alerts
Systems & Networking: Python, Bash, SQL, PostgreSQL, Linux, Git, TCP/IP, HTTP/HTTPS, DNS, BGP, MPLS

Projects

EKS Reliability Platform | EKS, Terraform, Helm, Python/FastAPI, RDS GitHub

- Provisioned a multi-tier AWS environment with Terraform, including public, private, and database subnets, managed EKS, RDS PostgreSQL, ECR, Secrets Manager, IRSA, and CloudWatch monitoring.
- Implemented six Prometheus availability and latency recording rules with multi-window burn-rate alerts against a 99.9% SLO, validated through five promtool tests.
- Load-tested the EKS-hosted backend with 18,819 requests and zero failures or pod restarts; HPA scaled the service from 2 to 6 replicas under peak load and returned it to baseline afterward.

GitOps Delivery Platform | EKS, Argo CD, Helm, Terraform, GitHub Actions GitHub

- Built an EKS GitOps platform from 15 reusable Terraform modules, with four Argo CD Applications reconciling dev, staging, prod, and observability namespaces under namespace-scoped RBAC, quotas, and network policies.
- Secured image promotion with a four-gate GitHub Actions workflow that validates Git and ECR provenance, authenticates through branch-scoped OIDC without stored AWS credentials, and delivers each promotion through a pull request.

Serverless Uptime Monitor | Lambda, Python, DynamoDB, Terraform Live | GitHub

- Built a serverless uptime monitor with scheduled Lambda checks and state-change alerts; added API throttling, dead-letter queues, point-in-time recovery, KMS encryption, CloudWatch monitoring, and X-Ray tracing.
- Designed a latest-status access pattern that reads 10 current records instead of scanning 86,400 retained records in the reference workload, keeping dashboard-read cost constant as history grows.
- Validated stateful alerting through an outage drill with one-second failure detection, dashboard visibility within 26 seconds, duplicate-page suppression, and recovery notification.

Experience

Tata Communications July 2022 – July 2024

Senior Engineer, Network Operations Center – Shift Lead Pune, India

Tier-1 global carrier; subsea and terrestrial backbone carrying roughly 30% of the world's internet routes

- Led five-engineer shifts in a 24/7 NOC, triaging and prioritizing 40+ daily incidents across 25–30 Tier-1 carrier clients under 99.9% availability and four-hour restoration commitments.
- Diagnosed multi-layer service outages by isolating faults across BGP/MPLS routing, TCP/IP connectivity, DWDM transport, and network hardware.
- Owned high-severity P1/P2 incident response end to end in ServiceNow, coordinating vendor TAC escalation, field dispatch, customer-premises testing, and client communication through restoration.
- Produced post-incident RCAs and structured handoffs that preserved diagnostic context across shifts; earned a Certificate of Excellence for resolving escalated customer cases.

Education

Indiana University Bloomington August 2024 – May 2026

Master of Science in Computer Science, GPA: 3.5/4.0 Bloomington, IN

SRM Institute of Science and Technology June 2018 – June 2022

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.43/10 Chennai, India